2026-2027 – Econ 0107 – Macroeconomics I

Chapter 5 : Dynamic Competitive Equilibrium

(Chapter 5 in Azzimonti, Krusell, McKay, Mukoyama and others's textbook)

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Version 1.0
19/10/2026

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1. Introduction

What Is a Competitive Equilibrium?

- Two conditions must hold simultaneously:
  1. Agents maximize their objectives subject to constraints, taking prices as given
  2. Markets clear: demand equals supply for all traded goods
- **Three formulations**:
  1. Arrow-Debreu: all trades at date 0; prices $\{p_t\}_{t=0}^{\infty}$
  2. **Sequential**: trades period by period; bond price q_t each period
  3. **Recursive**: prices and policies as functions of state variables
- For many economies, all three yield the **same allocations**. The first two express equilibria as sequences; the third as functions.

2. Arrow-Debreu Equilibrium

Endowment Economy: Primitives

- Continuum of consumers $i \in [0, 1]$; each produces $y_{i,t} \in \mathbb{R}_+$ of a single non-storable good
- Common preferences:

$$\sum_{t=0}^{\infty} \beta^t u(c_{i,t}),$$

  with u strictly increasing and strictly concave
- Price p_t: price of time-t good in terms of time-0 good; normalize $p_0 = 1$
- Budget constraint (equality since u strictly increasing):

$$\sum_{t=0}^{\infty} p_t c_{i,t} = \sum_{t=0}^{\infty} p_t y_{i,t}$$

- Contracts are fully enforcable

Definition 1

An ARROW-DEBREU *competitive equilibrium is a set of sequences* $\{c^\star_{i,t}\}_{t=0}^\infty$*, for each* $i \in [0,1]$*, and* $\{p_t\}_{t=0}^\infty$ *such that*

1. *for each* i, $\{c^\star_{i,t}\}_{t=0}^\infty$ *solves*

$$\max_{\{c_t\}_{t=0}^\infty} \sum_{t=0}^\infty \beta^t u(c_t) \quad s.t. \quad \sum_{t=0}^\infty p_t c_t = \sum_{t=0}^\infty p_t y_{i,t}$$

2. *for each* t,

$$\int_0^1 c^\star_{i,t}\, di = \int_0^1 y_{i,t}\, di$$

- Point 1: optimization given prices. Point 2: choosing prices such that goods market clears.

Characterization: Prices

- FOC from the Lagrangian on the lifetime budget:

$$\beta^t u'(c_{i,t}) = \lambda_i p_t$$

  where λ_i is the multiplier on the lifetime budget constraint. It does not depend on t.

- Eliminating λ_i between $t = 0$ and a generic t:

$$p_t = \beta^t \frac{u'(c_{i,t})}{u'(c_{i,0})}$$

- Since p_t is common to all agents: the ratio of marginal utilities at different dates is **equalized across all consumers**.

Log Utility: Prices and Consumption Growth

- With $u(\cdot) = \log(\cdot)$: $p_t = \beta^t c_{i,0}/c_{i,t}$. Multiply by $c_{i,t}$, integrate over i, use market clearing ($\int_0^1 c_{i,t}\, di = Y_t$):

$$p_t\, c_{i,t} = \beta^t c_{i,0} \;\Rightarrow\; p_t \int_0^1 c_{i,t}\, di = \beta^t \int_0^1 c_{i,0}\, di \;\Rightarrow\; p_t Y_t = \beta^t Y_0$$

$$p_t = \beta^t \frac{Y_0}{Y_t}$$

  p_t is low if t is large or if the endowment is large.

- All consumers share the same consumption growth:

$$\frac{c_{i,t+1}}{c_{i,t}} = \frac{Y_{t+1}}{Y_t} = \beta \frac{p_t}{p_{t+1}} \quad \text{for all } i$$

- If $Y_t = \bar{Y}$ constant: $c_{i,t}$ is constant for all i and t (**perfect consumption smoothing**).

- From the Euler equations and the budget constraint:

$$\underbrace{\sum_{t=0}^{\infty} p_t c_{i,t}}_{=\sum_t \beta^t c_{i,0} = \frac{c_{i,0}}{1-\beta}} = \underbrace{\sum_{t=0}^{\infty} p_t y_{i,t}}_{=Y_0 \sum_t \beta^t \frac{y_{i,t}}{Y_t}} \quad \text{using } p_t c_{i,t} = \beta^t c_{i,0} \text{ and } p_t = \beta^t \frac{Y_0}{Y_t}$$

- Hence:

$$c_{i,0} = (1-\beta) Y_0 \sum_{t=0}^{\infty} \beta^t \frac{y_{i,t}}{Y_t}$$

Consumption Shares

- Individual i's share of aggregate consumption is constant over time:

$$c_{i,t} = \beta^t \frac{c_{i,0}}{p_t} = \beta^t c_{i,0} \frac{Y_t}{\beta^t Y_0} = \frac{c_{i,0}}{Y_0} Y_t$$

$$c_{i,t} = \theta_i Y_t, \qquad \theta_i = (1-\beta) \sum_{t=0}^{\infty} \beta^t \frac{y_{i,t}}{Y_t}$$

- Endowments in periods with high Y_t receive lower weight (marginal utility is lower).
- Two consumers with the same average endowments can have different wealth: the wealthy consumer is the one whose endowments are high when others' endowments are low.
- **General CRRA**: $p_t = \beta^t (Y_0/Y_t)^\sigma$. Higher $\sigma \Rightarrow$ resources more valuable in low-endowment periods.

Definition 2

Definition 2 (Representative Agent)

When all agents have equal endowments and u is strictly concave, all consumption choices are identical:
An AD CE is $\{c_t^\star\}_{t=0}^\infty$ and $\{p_t\}_{t=0}^\infty$ such that

1. $\{c_t^\star\}_{t=0}^\infty$ *solves*

$$\max_{\{c_t\}_{t=0}^\infty} \sum_{t=0}^\infty \beta^t u(c_t) \quad s.t. \quad \sum_{t=0}^\infty p_t c_t = \sum_{t=0}^\infty p_t y_t$$

2. $c_t^\star = y_t$

Production Economy with Labor: Primitives

- Household i: efficiency units e_i, hours $\ell_{i,t}$, effective hours $e_i \ell_{i,t}$, utility:

$$\sum_{t=0}^{\infty} \beta^t \left[u(c_{i,t}) - v(\ell_{i,t}) \right],$$

  with v increasing and convex
- Firm: $Y_t = A_t L_t$ (CRS), hires labor at wage w_t per efficiency unit
- Firm profit max:

$$\max_{\{L_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \{ p_t A_t L_t - p_t w_t L_t \}$$

- Household budget:

$$\sum_{t=0}^{\infty} p_t c_{i,t} = \sum_{t=0}^{\infty} p_t w_t e_i \ell_{i,t}$$

Definition 3

Definition 3

An AD CE is $\{c^\star_{i,t}\}_{t=0}^\infty$, $\{\ell^\star_{i,t}\}_{t=0}^\infty$ *for each* $i \in [0,1]$, $\{L^\star_t\}_{t=0}^\infty$, $\{p_t\}_{t=0}^\infty$, $\{w_t\}_{t=0}^\infty$ *such that*

1. *for each* i, $\{c^\star_{i,t}\}_{t=0}^\infty$ *and* $\{\ell^\star_{i,t}\}_{t=0}^\infty$ *solve*

$$\max_{\{c_t,\ell_t\}_{t=0}^\infty} \sum_{t=0}^\infty \beta^t[u(c_t) - v(\ell_t)] \quad s.t. \quad \sum_{t=0}^\infty p_t c_t = \sum_{t=0}^\infty p_t w_t e_i \ell_t$$

2. $\{L^\star_t\}_{t=0}^\infty$ *solves*

$$\max_{\{L_t\}_{t=0}^\infty} \sum_{t=0}^\infty \{p_t A_t L_t - p_t w_t L_t\}$$

3. *for each* t, *good and labour market clear:*

$$\int_0^1 e_i \ell^\star_{i,t}\, di = L^\star_t \qquad \text{and} \qquad \int_0^1 c^\star_{i,t}\, di = A_t L^\star_t$$

- **Household FOCs**:

$$\beta^t u'(c_{i,t}) = \lambda_i p_t, \qquad v'(\ell_{i,t}) = e_i w_t u'(c_{i,t})$$

- **Firm FOC**: For an interior solution with CRS, $w_t = A_t$. If $A_t > w_t$: no finite optimum. If $A_t < w_t$: firm shuts down.
- With log utility and $w_t = A_t$: the intratemporal FOC becomes $v'(\ell_{i,t})c_{i,t} = e_i A_t$. With $c_{i,t} = \theta_i Y_t$ and conjecturing $\theta_i = e_i$: all agents supply the same labor $\ell_{i,t} = L_t$. Income and substitution effects exactly cancel.

With heterogeneous initial assets

- Individual i is endowed with $a_{i,0}$ at time zero

$$\int_0^1 a_{i,0}\, di = 0$$

- Agents with different asset holdings will work different amounts because assets generate an income effect but not a substitution effect
- Consider the simple static case
  - × Budget constraint

$$c_i = e_i A \ell_i + a_i$$

  - × First-order condition, be given by

$$v'(\ell_i) = a_i A / (e_i A \ell_i + a_i)$$

  - × Individual labor supply depends nonlinearly on a_i
  - × Total labor supply and total consumption, will depend on the distribution of assets (inequalities shape macro allocations)

Neoclassical Growth Economy: Primitives

- Production:

$$Y_t = A_t K_t^{\alpha} L_t^{1-\alpha}$$

- Capital:

$$k_{i,t+1} = (1-\delta)k_{i,t} + \iota_{i,t}$$

- Firm:

$$\max_{\{K_t, L_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \{p_t A_t K_t^{\alpha} L_t^{1-\alpha} - p_t r_t K_t - p_t w_t L_t\}$$

- Household budget:

$$\sum_{t=0}^{\infty} p_t(c_{i,t} + \iota_{i,t}) = \sum_{t=0}^{\infty} p_t(r_t k_{i,t} + w_t e_i \ell_{i,t})$$

Definition 4

Definition 4

An AD CE is $\{c_{i,t}^\star\}$, $\{\iota_{i,t}^\star\}$, $\{\ell_{i,t}^\star\}$ *for each* i, $\{L_t^\star\}$, $\{K_{t+1}^\star\}$, $\{p_t\}$, $\{p_t w_t\}$, $\{p_t r_t\}$ *such that*

1. *for each* i, $\{c_{i,t}^\star\}$, $\{\iota_{i,t}^\star\}$, $\{\ell_{i,t}^\star\}$ *solve*

$$\max_{\{c_t,\iota_t,\ell_t\}} \sum_{t=0}^{\infty} \beta^t [u(c_t) - v(\ell_t)]$$

$$s.t. \quad \sum_{t=0}^{\infty} p_t(c_t + \iota_t) = \sum_{t=0}^{\infty} p_t(r_t k_t + w_t e_i \ell_t)$$

   where $k_{t+1} = (1-\delta)k_t + \iota_t$, $k_0 = k_{i,0}$

2. $\{L_t^\star\}$, $\{K_t^\star\}$ *(with* $K_0^\star = \int_0^1 k_{i,0}\, di$*) solve*

$$\max_{\{K_t,L_t\}} \sum_{t=0}^{\infty} \{p_t A_t K_t^\alpha L_t^{1-\alpha} - p_t r_t K_t - p_t w_t L_t\}$$

3. $\int_0^1 e_i \ell_{i,t}^\star\, di = L_t^\star$, $\int_0^1 k_{i,t}^\star\, di = K_t^\star$, $\int_0^1 (c_{i,t}^\star + \iota_{i,t}^\star)\, di = A_t (K_t^\star)^\alpha (L_t^\star)^{1-\alpha}$

Optimality Conditions for Definition 4

- **Consumption and labor** (same structure as before):

$$\beta^t u'(c_{i,t}) = \lambda_i p_t, \qquad v'(\ell_{i,t}) = e_i w_t u'(c_{i,t})$$

- **Investment** (new): the household's FOC for $\iota_{i,t}$ yields

$$p_t = p_{t+1} r_{t+1} + (1-\delta) p_{t+1} = p_{t+1}[r_{t+1} + 1 - \delta]$$

  Cost of one unit of capital today = rental revenue + undepreciated value tomorrow.

- **Firm FOCs**: $r_t = \alpha A_t K_t^{\alpha-1} L_t^{1-\alpha}$, $w_t = (1-\alpha) A_t K_t^{\alpha} L_t^{-\alpha}$. CRS $\Rightarrow$ zero profits.

Equivalence with the Planner

- With identical agents ($e_i = 1$, $k_{i,0} = K_0$) and log utility, the aggregate equilibrium conditions become:

$$\frac{C_{t+1}}{C_t} = \beta[1 - \delta + \alpha A_{t+1} K_{t+1}^{\alpha-1} L_{t+1}^{1-\alpha}]$$

$$v'(L_t) = \frac{(1-\alpha) A_t K_t^{\alpha} L_t^{-\alpha}}{C_t}$$

$$C_t + K_{t+1} = A_t K_t^{\alpha} L_t^{1-\alpha} + (1-\delta) K_t$$

- These are identical to the FOCs of the social planner from Chapter 4 $\Rightarrow$ **First Welfare Theorem**.

3. Sequential Equilibrium

Sequential Trading: Setup

- Trades occur period by period; bond price q_t (pay q_t today, receive 1 tomorrow)
- Gross interest rate:
$$1 + r_{t+1} = 1/q_t$$
- Period budget:
$$c_{i,t} + q_t a_{i,t+1} = y_{i,t} + a_{i,t}$$
- Need a no-Ponzi-game condition to rule out infinite borrowing
- Initial assets:
$$\int_0^1 a_{i,0}\, di = 0$$

Definition 5

Definition 5

A sequential CE is $\{c_{i,t}^\star\}_{t=0}^\infty$, $\{a_{i,t+1}^\star\}_{t=0}^\infty$ *for each* $i \in [0,1]$, *and* $\{q_t\}_{t=0}^\infty$ *such that*

1. *for each* i, $\{c_{i,t}^\star\}_{t=0}^\infty$ *and* $\{a_{i,t+1}^\star\}_{t=0}^\infty$ *solve*

$$\max_{\{c_{i,t},a_{i,t+1}\}_{t=0}^\infty} \sum_{t=0}^{\infty} \beta^t u(c_{i,t})$$

$$s.t. \quad c_{i,t} + q_t a_{i,t+1} = a_{i,t} + y_{i,t} \text{ for all } t,$$

$$\lim_{t\to\infty} \left(\prod_{s=0}^{t} q_s \right) a_{i,t+1} \geq 0 \quad (nPg)$$

2. *for all* t,

$$\int_0^1 c_{i,t}^\star \, di = \int_0^1 y_{i,t} \, di \qquad \text{and} \qquad \int_0^1 a_{i,t+1}^\star \, di = 0$$

- The second clearing condition says aggregate assets net out to zero. By WALRAS' Law, only one clearing condition is needed.

Sequential FOCs

- FOCs from the Lagrangian with period-by-period budget constraints:

$$\beta^t u'(c_{i,t}) = \lambda_{i,t}, \qquad q_t \lambda_{i,t} = \lambda_{i,t+1}$$

- Unlike the AD case, $\lambda_{i,t}$ now depends on t (one multiplier per period budget).
- Eliminating multipliers:

$$q_t = \beta \frac{u'(c_{i,t+1})}{u'(c_{i,t})}$$

Equivalence: Sequential $\Leftrightarrow$ Arrow-Debreu

- Chaining from 0 to $t-1$:

$$\prod_{s=0}^{t-1} q_s = \prod_{s=0}^{t-1} \beta \frac{u'(c_{i,s+1})}{u'(c_{i,s})} = \beta^t \, \frac{u'(c_{i,1})}{u'(c_{i,0})} \cdot \frac{u'(c_{i,2})}{u'(c_{i,1})} \cdots \frac{u'(c_{i,t})}{u'(c_{i,t-1})}$$

$$\prod_{s=0}^{t-1} q_s = \beta^t \frac{u'(c_{i,t})}{u'(c_{i,0})} = p_t$$

  The product of sequential bond prices equals the Arrow-Debreu price. Also: $p_{t-1}/p_t = 1/q_{t-1} = 1 + r_t$.

- Consolidating period budgets forward and using TVC (so the asset term vanishes):

$$\sum_{t=0}^{\infty} p_t c_{i,t} = a_{i,0} + \sum_{t=0}^{\infty} p_t y_{i,t}$$

- Same lifetime budget $\Rightarrow$ same allocations.

Definition 6

Definition 6 (Sequential CE, NGM)

A sequential CE is $\{c^\star_{i,t}\}_{t=0}^{\infty}$, $\{k^\star_{i,t+1}\}_{t=0}^{\infty}$ *for each* $i \in [0,1]$, $\{r_t\}_{t=0}^{\infty}$, $\{w_t\}_{t=0}^{\infty}$ *such that*

1. *for each* i, $\{c^\star_{i,t}\}_{t=0}^{\infty}$ *and* $\{k^\star_{i,t+1}\}_{t=0}^{\infty}$ *solve*

$$\max_{\{c_{i,t},k_{i,t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_{i,t})$$

$$s.t. \quad c_{i,t} + k_{i,t+1} = (1 - \delta + r_t)k_{i,t} + w_t e_i \text{ for all } t,$$

$$\lim_{t\to\infty} \frac{k_{i,t+1}}{\prod_{s=0}^{t}(1-\delta+r_{s+1})} \geq 0 \quad (nPg)$$

2. *for each* t, $(K_t, 1)$ *solves* $\max_{K,L} A_t K^\alpha L^{1-\alpha} - r_t K - w_t L$, *with* $K_t = \int_0^1 k^\star_{i,t}\, di$
3. *for each* t, $C_t + K_{t+1} = A_t K_t^\alpha + (1-\delta)K_t$, *where* $C_t = \int_0^1 c^\star_{i,t}\, di$

- By Walras' Law, condition 3 is redundant given condition 2.

4. Recursive Equilibrium

Recursive Methods: Overview

- Express equilibrium as **functions** of state variables, not sequences indexed by time.
- A state variable must be both **relevant** (determines prices) and **predetermined**.
- Two steps:
  1. **Steady-state** recursive equilibria: aggregate variables constant; functions specify that agents *choose* to remain at constant levels
  2. **Dynamic** recursive equilibria: aggregates change with the state K; agents use K to forecast future prices via $r(K)$, $w(K)$, $K' = G(K)$

Definition 7

Definition 7 (Steady-State CE, Endowment Economy)

With constant endowments y_i for each i. A recursive steady-state CE is a q and a set of functions $V_i^\star(a)$ and $g_i^\star(a)$, and asset values $a_i^\star$, for each $i \in [0,1]$, such that

1. *for each i, $V_i^\star(a)$ solves*

$$V_i(a_i) = \max_{a_i'} \{u(a_i + y_i - q a_i') + \beta V_i(a_i')\} \quad \forall a_i$$

   with $g_i^\star(a_i)$ attaining the max for all a_i

2. *for each i, $g_i^\star(a_i^\star) = a_i^\star$ (stationarity)*

3. *$\int_0^1 g_i^\star(a_i^\star)\, di = 0$ (asset market clearing)*

Characterization of Definition 7

$$V_i(a_i) = \max_{a_i'} \left\{ u(a_i + y_i - qa_i') + \beta V_i(a_i') \right\} \quad \forall a$$

- FOC wrt a_i':
$$qu'(a_i + y_i - qa_i') = V'(a_i')$$
- Envelope condition
$$V'(a_{i)} = u'(a_i + y_i - qa_i')$$
which implies
$$V'(a_i)' = u'(a_i' + y_i - qa_i'')$$
- now use
$$a_i' = g_i(a_i)$$

Characterization of Definition 7

- The functional Euler equation (FOC + Envelope Theorem), for all i:

$$qu'(a_i + y_i - qg_i(a_i)) = \beta u'(g_i(a_i) + y_i - qg_i(g_i(a_i)))$$

- Evaluating at $a_i = a_i^\star$ with stationarity ($g_i^\star(a_i^\star) = a_i^\star$):

$$q = \beta$$

- **Policy**: $g_i^\star(a_i) = a_i$ for all i (permanent-income result in recursive form).
- **Value**: $V_i^\star(a_i) = u(a_i(1-q) + y_i)/(1-\beta)$.
- The asset distribution $\{a_i^\star\}$ is indeterminate subject to $\int_0^1 a_i^\star \, di = 0$. This model has no predictions for long-run wealth distributions (Chapter 21 addresses this).

Definition 8

Definition 8 (Steady-State CE, NGM)

A recursive steady-state CE consists of scalars r and w and a set of functions $V_i^\star(k_i)$ and $g_i^\star(k_i)$, and capital holdings $k_i^\star$, for each $i \in [0,1]$, such that

1. *for each i, $V_i^\star(k_i)$ solves*

$$V_i(k_i) = \max_{k_i'} \{u((1-\delta+r)k_i + we_i - k_i') + \beta V_i(k_i')\} \quad \forall k_i$$

   with $g_i^\star(k_i)$ attaining the max for all k_i

2. *$(K^\star, 1)$ solves*

$$\max_{K,L} F(K,L) - rK - wL, \quad \text{where} \quad \int_0^1 k_i^\star \, di = K^\star$$

3. *for each i, $g_i^\star(k_i^\star) = k_i^\star$ (stationarity)*

Characterization of Definition 8

- From the Euler equation at stationarity:

$$\beta = 1/(1 - \delta + r),$$

  which pins down r.
- Then $r = F_1(K^\star, 1)$ determines $K^\star$, and $w = F_2(K^\star, 1)$ determines w.
- **Policies**: $g_i^\star(k_i) = k_i$ and

$$V_i^\star(k_i) = u(k_i(r - \delta) + we_i)/(1 - \beta)$$

- The distribution of capital is indeterminate, subject to

$$\int_0^1 k_i^\star \, di = K^\star.$$

Dynamic Recursive CE: Setup

- Out of steady state: prices depend on aggregate capital K (relevant and predetermined).
- Agent takes as given: price functions $r(K)$, $w(K)$, and a law of motion $K' = G(K)$.
- Individual Bellman equation (must allow $k_i \neq K$):

$$V(k_i, K) = \max_{k_i'} \left\{ u((1 - \delta + r(K))k_i + w(K) - k_i') + \beta V(k_i', G(K)) \right\}$$

for all (k_i, K). Agent tracks K because prices depend on it.

Definition 9

Definition 9 (Recursive CE, NGM)

A recursive CE consists of functions $r(K)$, $w(K)$, $G^\star(K)$, $V^\star(k_i, K)$, and $g^\star(k_i, K)$ such that

1. $V^\star(k_i, K)$ *solves*

$$V(k_i, K) = \max_{k_i'} \{u((1 - \delta + r(K))k_i + w(K) - k_i') + \beta V(k_i', G^\star(K))\}$$

   for all (k_i, K), with $g^\star(k_i, K)$ attaining the max

2. $r(K) = F_1(K, 1)$ *and* $w(K) = F_2(K, 1)$ *for all* K
3. $G^\star(K) = g^\star(K, K)$ *for all* K *(consistency)*

- Consistency: aggregate capital evolves according to individual choices when each agent holds $k_i = K$. The resource constraint is automatically satisfied.

Functional Euler Equation and Equivalence

- FOC + Envelope Theorem, evaluated at $k_i = K$:

$$u'(F(K,1) + (1-\delta)K - G(K)) =$$

$$\beta u'(F(G(K),1) + (1-\delta)G(K) - G(G(K))) \cdot [1 - \delta + F_1(G(K),1)]$$

- This functional equation in G coincides with the planner's functional Euler equation from Chapter 4. Must be solved numerically unless $u = \log$, F Cobb-Douglas, $\delta = 1$.

Aggregation

- With power utility, individual saving takes the form (proof on slide 38):

$$g(k_i, K) = \mu(K) + \lambda(K) k_i$$

- Saving is linear in own capital; slope $\lambda(K)$ is independent of k
- Wealth distribution does not affect aggregate saving $\Rightarrow$ not restrictive to use a representative agent
- With heterogeneous e_i: $g_i(k, K) = \mu_i(K) + \lambda(K)k$, with $\mu_i(K) = e_i\,\mu(K)$ (intercept proportional to e_i, slope common) (proof on slide 39) $\rightsquigarrow$ again, not restrictive to use a representative agent

Aggregation: Proof (Guess and Verify)

- $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$, $R(K) \equiv 1 - \delta + r(K)$, $K' = G(K)$. Euler equation:

$$c(k_i, K)^{-\sigma} = \beta R(K')\, c(k_i', K')^{-\sigma} \;\Leftrightarrow\; c(k_i', K') = \left(\beta R(K')\right)^{1/\sigma} c(k_i, K)$$

- Guess consumption linear in own capital: $c(k_i, K) = a(K) + b(K) k_i$. The budget gives

$$k_i' = R(K) k_i + w(K) - c(k_i, K) = \underbrace{w(K) - a(K)}_{\mu(K)} + \underbrace{\left(R(K) - b(K)\right)}_{\lambda(K)} k_i$$

- Plug into the Euler equation; both sides are linear in k:

$$a(K') + b(K')\left[\mu(K) + \lambda(K) k_i\right] = \left(\beta R(K')\right)^{1/\sigma} \left[a(K) + b(K) k_i\right]$$

- Match coefficients (for all k_i):

$$\begin{aligned} k: &\quad b(K')\left(R(K) - b(K)\right) = \left(\beta R(K')\right)^{1/\sigma} b(K) \\ 1: &\quad a(K') + b(K')\left(w(K) - a(K)\right) = \left(\beta R(K')\right)^{1/\sigma} a(K) \end{aligned}$$

- These functional equations in $a(\cdot)$, $b(\cdot)$ involve K only, not k_i $\Rightarrow$ the guess is verified, and $g(k_i, K) = \mu(K) + \lambda(K) k$

Aggregation: Proof with Heterogeneous e_i

- Budget of agent i: $k_i' = R(K)k_i + w(K)e_i - c_i(k_i, K)$, with $\int_0^1 e_i\, di = 1$
- Guess $c_i(k_i, K) = a_i(K) + b_i(K)k_i$. Same steps as before give, for each i:

$$\begin{aligned} k_i: &\quad b_i(K')\left(R(K) - b_i(K)\right) = \left(\beta R(K')\right)^{1/\sigma} b_i(K) \\ 1: &\quad a_i(K') + b_i(K')\left(w(K)e_i - a_i(K)\right) = \left(\beta R(K')\right)^{1/\sigma} a_i(K) \end{aligned}$$

- The k_i-equation does not involve $e_i \Rightarrow$ same solution for all i: $b_i = b$, so $\lambda(K) = R(K) - b(K)$ is common
- The 1-equation is linear and homogeneous in $(a_i, e_i) \Rightarrow a_i(K) = e_i\, a(K)$, where a solves the equation with $e_i = 1$
- Hence

$$g_i(k_i, K) = \underbrace{e_i\left(w(K) - a(K)\right)}_{\mu_i(K) = e_i \mu(K)} + \lambda(K)k_i$$

- Aggregating: $\int_0^1 g_i(k_i, K)\, di = \mu(K)\int_0^1 e_i\, di + \lambda(K)\int_0^1 k_i\, di = \mu(K) + \lambda(K)K = G(K)$

Definition 10

- Assume agents have utility functions $u(c) - v(\ell)$

Definition 10 (Recursive CE with Valued Leisure)

A recursive CE consists of $r(K)$, $w(K)$, $G^\star(K)$, $H^\star(K)$, $V^\star(k_i, K)$, $g^\star(k_i, K)$, $h^\star(k_i, K)$ such that

1. $V^\star(k_i, K)$ *solves*

$$V(k_i, K) = \max_{k_i', \ell} \{u((1 - \delta + r(K))k_i + w(K)\ell - k_i') - v(\ell) + \beta V(k_i', G^\star(K))\}$$

   for all (k_i, K), with $g^\star$, $h^\star$ attaining the max

2. $r(K) = F_1(K, H^\star(K))$ *and* $w(K) = F_2(K, H^\star(K))$ *for all* K
3. $G^\star(K) = g^\star(K, K)$ *and* $H^\star(K) = h^\star(K, K)$ *for all* K

- The agent does not use $H^\star$ directly; $G^\star$ suffices because r and w capture how labor input changes with K. Aggregation breaks unless $k_{i,0}/e_i$ is equalized across agents.

5. Overlapping Generations

OLG: Setup

- Agents live two periods (y = young, o = old); no altruism, no bequests
- Cohort t utility: $u_t(c_y, c_o) = u(c_y) + \beta u(c_o)$
- Initial old ($t = -1$): $u_{-1}(c) = u(c)$
- Endowments: $(\omega_{y,t}, \omega_{o,t+1})$ for cohort t
- **Key differences from dynastic models**:
  - × Simpler maximization (finite horizon)
  - × First-order dynamics (not second-order)
  - × Equilibria can be Pareto inefficient
  - × Negative real interest rates possible

Definition 11

Definition 11 (Sequential CE, OLG Endowment)

A sequential CE is $\{c^\star_{i,t}\}_{t=0}^{\infty}$ *for each* $i \in \{y, o\}$, $\{a^\star_{t+1}\}_{t=0}^{\infty}$, *and* $\{q_t\}_{t=0}^{\infty}$ *such that*

1. *for each* $t > 0$, $(c^\star_{y,t}, c^\star_{o,t+1}, a^\star_{t+1})$ *solves*

$$\max_{c_y, c_o, a'} u(c_y) + \beta u(c_o)$$

$$\text{s.t.} \quad c_y + q_t a' = \omega_{y,t} \quad \text{and} \quad c_o = \omega_{o,t+1} + a'$$

   and $c^\star_{o,0} = \omega_{o,0}$

2. *for all* $t \geq 0$, $c^\star_{y,t} + c^\star_{o,t} = \omega_{y,t} + \omega_{o,t}$

- Goods market clearing $\Leftrightarrow$ $a^\star_{t+1} = 0$ for all t: any saving by the young requires someone on the other side, but the old are in their last period and the next cohort is unborn.

Characterization of Definition 11

- Since $a^\star_{t+1} = 0$: **autarky**, $(c^\star_{y,t}, c^\star_{o,t}) = (\omega_{y,t}, \omega_{o,t})$.
- Equilibrium price from the Euler equation at autarky (CRRA utility):

$$q_t = \beta \left(\frac{\omega_{y,t}}{\omega_{o,t+1}} \right)^\sigma$$

- If $\omega_y > \omega_o$ (income declines over the life cycle, but not with time): $q > 1$ possible $\Rightarrow$ **negative real interest rate**. Cannot happen in the dynastic model (unless aggregate endowments decline with time).
- **Intuition**: No smoothing feasible (old cannot trade with unborn). Transferring from young to old (e.g. PAYG pension) can make all generations better off $\Rightarrow$ market outcome is inefficient. Chapter 6 analyzes this.

Definition 12

Definition 12 (Sequential CE, OLG Growth Economy)

Labor productivities e_y, e_o with $e_y + e_o = 1$; stationary $F(k, \ell)$.
A sequential CE is $\{c^\star_{i,t}\}_{t=0}^\infty$ for each $i \in \{y, o\}$, $\{k^\star_{t+1}\}_{t=0}^\infty$, $\{r_t\}_{t=0}^\infty$, $\{w_t\}_{t=0}^\infty$ such that

1. *for each $t > 0$, $(c^\star_{y,t}, c^\star_{o,t+1}, k^\star_{t+1})$ solves*

$$\max_{c_y, c_o, k'} u(c_y) + \beta u(c_o)$$

$$\text{s.t.} \quad c_y + k' = w_t e_y \quad \text{and} \quad c_o = w_{t+1} e_o + (1 - \delta + r_{t+1}) k'$$

   and $c^\star_{o,0} = w_0 e_o + (1 - \delta + r_0) k_0$

2. *for all t, $r_t = F_1(k^\star_t, 1)$ and $w_t = F_2(k^\star_t, 1)$, with $k^\star_0 \equiv k_0$*
3. *for all $t \geq 0$, $c^\star_{y,t} + c^\star_{o,t} + k^\star_{t+1} = F(k^\star_t, 1) + (1 - \delta) k^\star_t$*

OLG Euler Equation and Dynamics

- Substituting prices as functions of capital:

$$u'(e_y F_2(k_t, 1) - k_{t+1}) =$$

$$\beta u'(e_o F_2(k_{t+1}, 1) + (1 - \delta + F_1(k_{t+1}, 1))k_{t+1}) \cdot (1 - \delta + F_1(k_{t+1}, 1))$$

- **First-order** difference equation: given k_t, solve for k_{t+1} and iterate forward
- Dynastic model: second-order (needs two boundary conditions)
- In general, $\beta(1 - \delta + F_1(\bar{k}, 1)) \neq 1$: the interest rate depends on life-cycle income
- No immediate connection to a planner's problem $\Rightarrow$ uniqueness not guaranteed

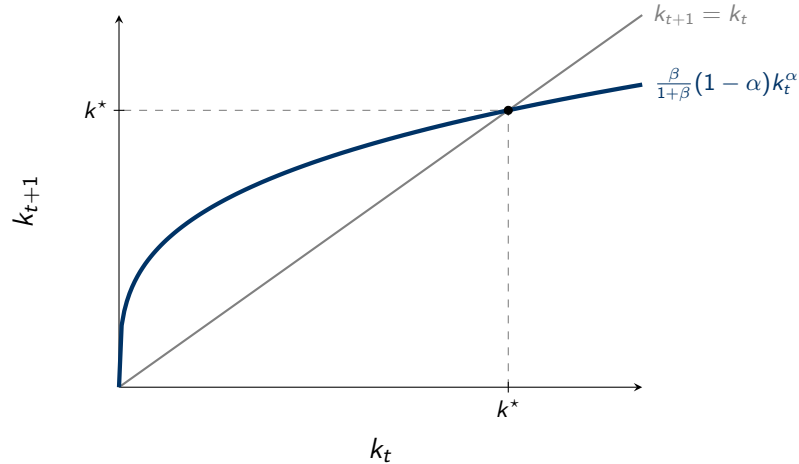
OLG: Example

- With $u = \log$, $e_y = 1$, $e_o = 0$, $F = K^\alpha L^{1-\alpha}$, $\delta = 1$:

$$k_{t+1} = \frac{\beta}{1+\beta}(1-\alpha)(k_t)^\alpha$$

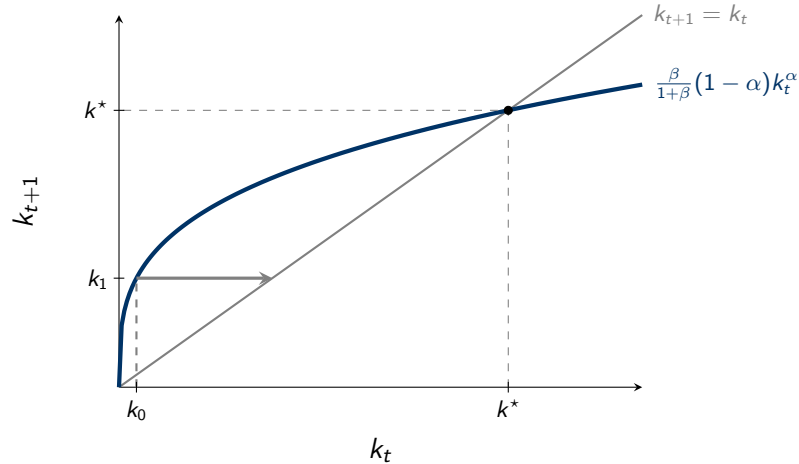
- Log-linear dynamics, monotonic convergence to steady state
- Steady-state gross interest rate: $\alpha(1+\beta)/(1-\alpha)$, which can be < 1 if α is low enough

OLG: Dynamics



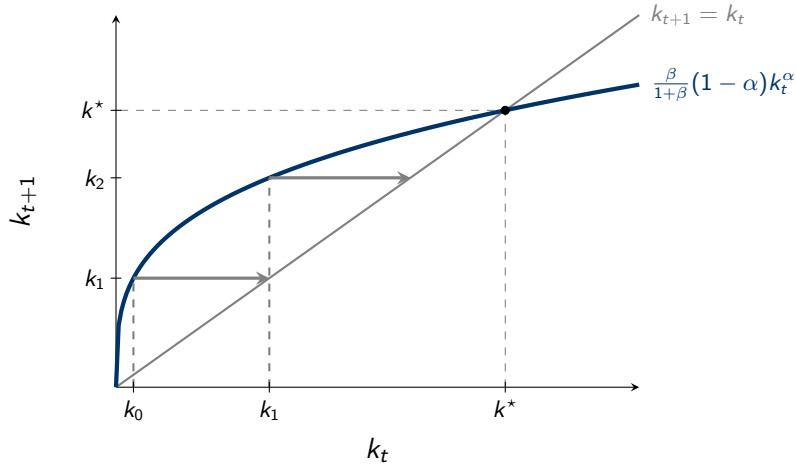
Example with $\alpha = 0.3$, $\beta = 1$: starting from $k_0 < k^\star$, capital rises monotonically to $k^\star = \left[\frac{\beta}{1+\beta}(1-\alpha)\right]^{\frac{1}{1-\alpha}}$.

OLG: Dynamics



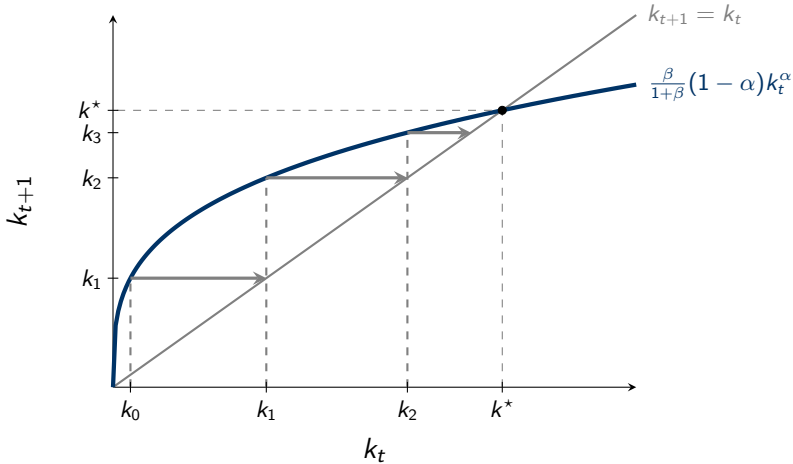
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OLG: Dynamics



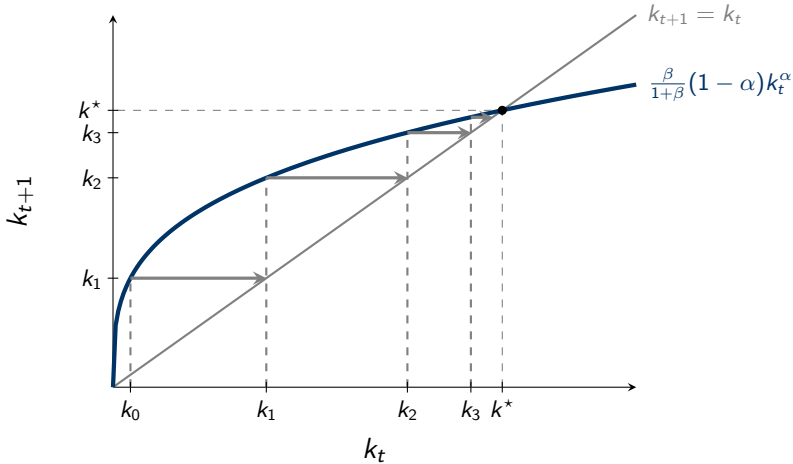
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OLG: Dynamics



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OLG: Dynamics



Example with $\alpha = 0.3$, $\beta = 1$: starting from $k_0 < k^\star$, capital rises monotonically to $k^\star = \left[\frac{\beta}{1+\beta}(1-\alpha)\right]^{\frac{1}{1-\alpha}}$.

- **Perpetual youth**: All individuals face a constant probability of death each period. As survival probability $\to 1$: approaches the dynastic model.
- **Warm glow**: $u(c_y) + \beta u(c_o) + \varphi(b')$, where φ is utility from the act of giving (behavioral model).
- **Pure altruism**: Parent values child's utility as the child does:

$$\varphi(b) = \max_{c_y, c_o, a', b'} \left\{ u(c_y) + \beta u(c_o) + \tilde{\beta} \varphi(b') \right\}$$

subject to budget constraints. Recursive structure $\Rightarrow$ dynastic life-cycle model with long-run interest rate $= 1/\tilde{\beta}$.

6. Summary

Key Takeaways (I)

1. **Three formulations**: Arrow-Debreu (Defs 1–4), sequential (Defs 5–6, 11–12), recursive (Defs 7–10). All yield the same allocations in dynastic economies.
2. **First Welfare Theorem**: With identical agents and no frictions, CE = planner's solution.
3. **Aggregation**: Power utility $\Rightarrow g(k, K) = \mu(K) + \lambda(K)k \Rightarrow$ wealth distribution irrelevant. Breaks with valued leisure.
4. **Investment optimality**: $p_t = p_{t+1}[r_{t+1} + 1 - \delta]$.

Key Takeaways (II)

5. **OLG**: First-order dynamics; interest rate depends on life-cycle income; negative rates possible; can be Pareto inefficient.
6. **Pure altruism**: OLG $\rightarrow$ dynasty as special case; long-run $r = 1/\tilde{\beta}$.
7. **Departures from perfect competition** (frictions, monopoly, incomplete markets) $\Rightarrow$ CE $\neq$ planner's solution. Chapters 6 and beyond.

